DataForce — build report (M1 to M4)

Status (19 October 2026):

Milestone	What it added	State
M1 ("jalon 1")	Data, CRM and statistics engine	complete and ratified
M1b	A local web interface, "Connect TamTam" over MCP, the 350-deal demo	complete, reviewed and fixed
M2a	Your feedback: drag-and-drop **analysis flows** with a pivot, statistics and model blocks; a **Companies** tab with the public French registry; an **AI model** in Settings with an **Ask** panel; a **light / dark** theme	complete, reviewed and fixed
M2b	Job titles normalised; org-chart **suggestions** from titles and from what people said in meetings (a person accepts them); **meeting transcripts** with known sales themes and discovered topics	complete, reviewed (security and methods) and fixed
M3a	An **MCP server**: Claude Desktop, Claude Code or another AI assistant on this computer queries one workspace read-only and gets exact results. Three **independent transcript tests**. An **opt-in AI pass** for org statements in meetings	complete, reviewed (SR14, SR15) and fixed; the AI pass awaits a live measurement
UI identity	One look for the whole app: Helvetica, colour and spacing **tokens**, **light and dark**; an assistant drawer, a deal stage roadmap, data-quality bars, a columns grid and honest node states on the flow canvas	complete, reviewed (SR16, 0 high or medium, 3 low) and fixed
Code audit and quick wins	A board of five reviewers measured unused code, speed and system risks; unused code removed; indexes, compression and faster queries added	complete; the ranked list is in `docs/review/CODE-AUDIT.md`; the structural items wait for the UI rework and M3b
Admin page	Database size, AI usage, recent runs and the real limits, read-only	complete
Company-wide assistant context	Ask on an account or a deal now sees the whole company's CRM data; a minimal deal page; a relaxed meeting AI pass (a manager named earlier in the meeting counts)	complete, reviewed (SR17, 3 medium, 4 low, 1 info) and fixed; reports-to on held-out set 4: 11 of 16 found, all correct

M4	The analytics suite: **activities and a timeline** on deals; **new flow blocks**; **workbooks** (visual canvases); **saved models and scoring**; **opt-in AI** on deals; **Gmail / Outlook mailbox** linking; a **methods review**	built and reviewed (SR18, SR19, M4-METHODS) and fixed; the mailbox awaits your OAuth clients

Every part was built by Sonnet agents, independently reviewed for security, fixed and re-verified. **1,772 automated tests pass** (1,608 backend, 164 frontend).

Sources of truth.

File	What it holds
`docs/DECISIONS.md`	every decision and its reason
`docs/ARCHITECTURE.md`	the M1 contract
`docs/UI-CONTRACT.md`	the M1b contract
`docs/M2A-CONTRACT.md`	the M2a contract
`docs/M2B-CONTRACT.md`	the M2b contract
`docs/review/M2b-methods-review.md`	the independent, held-out check of the M2b methods
`docs/M3A-CONTRACT.md`	the M3a contract (MCP server and agent tools)
`docs/ORG-AI-CONTRACT.md`	the contract of the opt-in AI pass on meeting sentences
`docs/M4-ANALYTICS-CONTRACT.md`	the M4 contract (analytics suite), with your decisions
`docs/review/M4-METHODS.md`	the independent methods review of the M4 blocks, with its resolution
`docs/review/CODE-AUDIT.md`	the code audit: unused code, speed, system risks, ranked
`docs/security/SR16` to `SR19`	the security reviews of the UI identity, the company-wide context, M4 phase 1 and M4 phase 2
`docs/UI-IDENTITY-CONTRACT.md`, `docs/VISUAL-IDENTITY.md`	the look of the app and its tokens
`docs/MAILBOX-SETUP.md`	how to register your Gmail and Outlook OAuth clients
`backend/eval/meetings/`	the independently written transcript sets, their answer keys and the scorer
`docs/BUILD-REPORT.md`	the source of this PDF, rendered by `docs/tools/build_pdf.py`

1. In one page

What DataForce is. An analysis workspace for sales and marketing teams. The team keeps a *minimal CRM* in DataForce: companies, accounts, people, deals and notes. Then it:

- looks up any company: "have we already dealt with them?";

- builds analyses by wiring blocks on a canvas;
- asks an AI about one account, with the AI's answers checked against the data.

TamTam already has better external data than we could build. So DataForce works at the CRM level, and calls TamTam only when a user explicitly asks for contact details.

The one rule that shapes everything. DataForce shows a number as a fact only when the evidence is strong enough. Otherwise it says **"not verified"** and gives the reasons:

- too few deals;
- an interval that is too wide;
- not significant once many comparisons are accounted for;
- for a model: not better than guessing, or tested on its own training data.

The AI never computes numbers. Any number it writes that is not in the data is flagged.

What is delivered.

Piece	What it does
Minimal CRM	Workspaces, companies grouping accounts, contacts with "reports to", deals, append-only notes and comments, one-click erasure
Companies (M2a)	One search box: have we interacted, and when last? A 360 page per company. On request, a lookup in the public French registry (address, SIREN, activity, size), with "Add to CRM"
Analysis flows (M2a)	Drag-and-drop canvas. Datasets come from the CRM, an uploaded CSV / Excel file or a read-only PostgreSQL database. Blocks: filter, sort, columns, group, join, **pivot**, stats test, chart, train/test split, linear and logistic regression, k-means
Statistics engine	Win rate, cycle length and deal size with 95 % confidence intervals and the "not verified" gates, reused by the stats and model blocks
AI model and Ask (M2a)	Gemini or Anthropic key entered in Settings, stored encrypted. The Ask panel on an account or company answers from its data, cites it, and flags numbers and quotes it cannot back
Job titles (M2b)	Every title mapped to a function and a seniority by a French / English dictionary; unknown titles optionally classified once by your AI model (title text only), flagged for review
Org-chart suggestions (M2b)	Managers proposed from titles (same account, next level up) and from meeting sentences ("I report to…"), plus champion / blocker / decision-maker signals, each with its evidence; nothing becomes a fact until a person accepts it
Meetings and themes (M2b)	Transcripts (PDF, Markdown, text) on accounts and deals; known sales themes found by keywords with exact quotes, other recurring topics discovered; "Themes vs win rate" as a screening flow. All local

Activities and timeline (M4)	A rep logs calls, emails and meetings on a deal; moving a deal to a stage takes a date and is recorded. The deal and account pages show one timeline with stage changes, activities, tasks, notes, comments and meeting titles
Analysis tabs (M4)	**Flows**, **Workbooks**, **Models** and **Insights**; new blocks (duplicates, aging, weighted forecast, window / QoQ, domain, SVM, deal velocity, engagement, touchpoint ratio, score)
AI on deals (M4)	Opt-in *Summarize* and *Suggest tasks*, after a dialog that shows what is sent
Mailbox (M4)	Read-only Gmail and Outlook linking of emails to open deals
Admin and data quality	An Admin page; health bars that count exactly what is missing
Connect TamTam (M1b)	Sign in on TamTam's page. The token is encrypted; paid lookups happen only after a confirmed click
Synthetic demo	350 deals with planted effects (seed 42) and a pure-noise twin (seed 1), with companies, a year of dated notes, half the org charts left to rediscover, and 160 meetings with planted themes
Command line and notebook	`dataforce …`; a proof notebook run against the answer key

Not delivered yet (by plan).

- M3b: login and hosting.
- The big UI rework you asked for after your UI report.
- The mailbox is built but not connected: it needs your OAuth clients.

2. How it was built

The team works the same way for every milestone:

1. The tech lead (Claude Opus) writes the contracts first: the database model, the shared data types, the API shapes, and a written specification with non-overlapping file ownership.
2. Builder agents (Claude Sonnet, low effort, as requested) work in parallel.
3. Separate reviewer agents can read but not change the code, and must prove each finding.
4. Fixers close the findings with regression tests.
5. The tech lead re-runs everything, opens every screen in a headless browser, and commits in logical steps.

Round	Build	Ratify (read-only)	Fix
M1	foundation, generator, stats engine, LLM clean-up, integration	SR4, SR5, statistics ratification	3 fixers
M1b	API and CRM, TamTam connector, frontend, 350-deal demo	SR6 (API), SR7 (OAuth), SR8 (frontend)	2 fixers + tech lead

M2a	flow engine, flow API and data sources, companies and registry, AI assistant, flow-editor UI, companies/AI/theme UI	SR9 (flows and data), SR10 (companies and AI), SR11 (frontend)	3 fixers + tech lead
M2b	titles, org proposals, transcripts and themes, demo meetings, org/titles UI, meetings UI; then a second and third theme method and realistic demo transcripts	SR12 (backend), SR13 (frontend), an **independent methods review** on held-out data	2 fixers + tech lead
UI identity, audit, SR17	tokens and shell, the five-reviewer audit, company-wide context, relaxed AI pass	SR16, SR17, the audit board	fixes by the tech lead
M4	contract, activities, blocks, workbooks, saved models, AI, mailbox	SR18, SR19, an independent **methods review**	fixes by a developer, checked by the tech lead

During M2a and M2b the agents hit usage limits mid-build. They were resumed where they stopped once the limit reset, with no work lost.

Version control. Logical commits on `main`, with the tags `m1`, `m1b`, `m2a`, `m2b` and `m3a`, in the private GitHub repository `nathnath2006/DataForce`; new commits are pushed when you ask.

3. Product decisions (and why)

Decision	Why
Not enough evidence → "not verified", always with the reasons; "not verified" is *not* "no effect".	A salesperson must be able to rely on whatever is shown as a fact.
The AI never calculates. Numbers come from code; the AI may only name, explain and cite them.	Numbers must be reproducible and auditable.
Notes and comments are append-only (a database trigger enforces it), except GDPR erasure.	Trust in what was said about a person, and the right to erasure.
Strict minimum on third parties. Emails and mobiles exist only after a confirmed TamTam lookup, encrypted, each with a *Delete*.	GDPR data minimisation.
Analysis is a flow of blocks (M2a, your request), not a fixed report.	Teams ask different questions; a pivot or a model should be two clicks away.
Blocks are configured, never programmed. There is no formula language, nothing is evaluated as code, and no model is saved.	A formula language and saved models are two classic ways in for attackers.
"Not verified" extends to models. A model is verified only if, on data it never saw, it beats the "learnt nothing" baseline with its confidence interval.	Most models on small CRM data are noise. The label says so.
External data is read-only. PostgreSQL runs one statement in a read-only transaction; files are copied into DataForce.	DataForce must never be able to change your systems.
A company groups accounts (M2a).	One customer often appears as several accounts: branches, subsidiaries.
The public registry is searched only when you click. The managers' list is never read; protected records are never stored.	Your search text leaves the computer only when you choose.
The AI key is entered in Settings, encrypted and never shown back.	Simple for you. The old "never paste the key" rule meant never paste it *into a chat*.
The Ask panel says what it sends (CRM names, titles, deals, notes; never emails or mobiles) and stores nothing.	You decide what goes to Google or Anthropic.
Local only, no login yet.	Login and hosting are M3.
Titles, links and themes: dictionary first (M2b). Code with fixed French / English vocabularies does the work; the AI is used only for what the dictionary does not know, and is flagged.	Explainable, cheap, private. The same pattern as the statistics: code computes, the AI names.
Suggestions are never facts (M2b). A rule or a meeting may only *suggest* a manager or a signal; a person accepts or rejects it.	An org chart is about people; a wrong "reports to" must never appear as certain.

Meeting text stays on the computer (M2b). Themes are found locally; the AI only names discovered topics, from top words and short excerpts with names masked.	Transcripts are the most sensitive data in the tool.
Erasing a person erases them from transcripts too (M2b). Their turns and their name are replaced by same-length placeholders.	GDPR erasure must not leave their words behind.
Opt-in AI on deals; a person accepts every suggested task (M4). Summaries cite real items and are not stored.	The AI may help read a deal, but never changes it alone.
The mailbox is read-only, stores only candidate emails, and you register your own OAuth clients (M4).	Minimal data, and no key of ours sits between you and your mail.
Scoring is "run now", no scheduler (M4).	A scheduler comes with hosting (M3b).
Methods review before trust (M4). An independent read-only review tested the new blocks on synthetic data and the fixes were measured.	Same rule as M2b: do not trust our own accuracy claims.
350 demo deals; noise demo on seed 1.	Your choices. Seed 42 of the noise twin is one of the ~4 % that show a chance finding.

4. How the statistics work (plain language)

The three questions.

- *Win rate*: of the closed deals, what share was won?
- *Sales cycle*: days from creation to close, as a median.
- *Deal size*: the median amount.

Medians are used so one giant deal does not distort the picture.

Confidence intervals. Every number comes with the range that, 95 times out of 100, contains the true value.

- Rates use Wilson intervals.
- Differences between two rates use Newcombe intervals.
- Medians use a bootstrap with a fixed seed, so results are reproducible.

The "not verified" gates (approved by you, all configurable):

A result is "not verified" when…	Default
the group has too few deals	fewer than 30
a win rate rests on too few wins or losses	fewer than 10 of either
the range is too wide	more than ±15 points for a rate, ±30 % of the value for a median

Comparing groups.

- Each group is compared with the rest, using Fisher's exact test for rates and Mann-Whitney for days and amounts.
- The Benjamini-Hochberg correction keeps the expected share of false findings at 5 %.

- A finding is **verified** only if both sides pass the gates, the corrected test is significant, and the range of the difference excludes zero.

Real lines from the demo (350 deals, seed 42, the same in the stats block of a flow):

Group vs rest	Question	Result	Verdict
Inbound leads	win rate	72.5 % vs 24.2 %, +48.4 pts [+35.9, +58.6]	**verified**
Enterprise	sales cycle	median 106 vs 41 days	**verified**
Referral leads	win rate	72.0 % vs 39.1 %	not verified: only 25 deals (< 30), only 7 losses
Mid-market	deal size	€20,771 vs €15,879	not verified: q = 0.58, range [−190, +9,921] EUR includes 0

Models (M2a). A split block keeps a share of the rows aside (25 % by default). The model learns on the rest and is judged only on the rows it never saw.

Model	Verified only if, on the held-out rows	Example on the demo
Logistic ("will this deal be won?")	at least 30 rows; enough wins and losses; AUC interval above 0.5; accuracy interval above the "always predict the majority" rate	accuracy 81.5 % [69.2, 89.6] vs baseline 53.7 %; AUC 0.86 [0.74, 0.97]: **verified**
Linear ("how large will it be?")	at least 30 rows; R^2 interval above 0	R^2 0.65 [0.46, …] on amount by segment: **verified**
Clustering (k-means)	never: groups describe, they do not prove	3 groups of accounts, silhouette 0.43: **exploratory**

Two traps are caught automatically.

- Wiring the same rows into both "train" and "test" gives "not verified: test rows may overlap the training rows".
- A join that duplicates rows (one row per contact instead of per deal) gives "not verified: n is inflated".

Coefficients are labelled "associations, not causes".

4b. Titles, org chart and meetings (M2b, plain language)

Titles.

- "Dir. commercial", "VP Sales" and "Ingénieur d'affaires" become *sales · head*, *sales · vp* and *sales · individual*.
- The dictionary expands abbreviations and repairs typos.
- A title it does not know is either sent once to your AI model (only the title, never the person's name) and flagged "classified by AI", or left "unknown" for you to set.

Org chart. For contacts without a manager, the rules look for the next level up in the same function, then the CEO, inside the same account. When two people fit, the rules list both and do not guess. Meeting sentences such as "I report to Soline Pellegrand" or "c'est Quentin qui signe" become suggestions, with the exact sentence as evidence.

Themes. We tried pure topic discovery (TF-IDF + NMF) first, and it failed on realistic transcripts: it returned small talk ("next week", "send the documents") and people's names.

The working method is the same as for titles:

1. A French / English catalog of common sales themes (price, competitor, security, integration, timing, legal, onboarding, support) is matched sentence by sentence, with exact quotes.
2. Competitor names are spotted when they recur near words like "similar offer".
3. Discovery then looks only at the remaining sentences. Its topics stay *exploratory*.

How good is it? There are two kinds of numbers:

- **In-sample**: on the synthetic demo, whose vocabulary the dictionary shares, so these flatter;
- **Held-out**: from an independent reviewer's own titles, sentences and org charts.

Method	Demo (in-sample)	Held-out	Verdict
Titles	99.6 % function, 99.7 % seniority	75.8 % function, 86.3 % seniority; 41 % left unknown or flagged, but **90 of 90 right when it answers**	ratified with caveats
Org rules	95 % precision, 81 % recall	88.5 % precision, 59 % recall; abstains rather than guesses	ratified with caveats
Meeting links and signals	100 % precision, 44 % recall; signals 74 of 74	narrow on paraphrases; a few false positives, fixed	ratified with caveats
Known themes	F1 0.82–0.98	precision 0.64–0.89, recall 0.35–0.70	**exploratory only**: no accuracy claim

The review's fixes were applied afterwards, so its held-out set has now been seen. A fresh one is needed for the next measurement.

Independent transcript tests (18 October). Each set holds 15 French and English meetings, written by a separate agent that saw only a brief of the accounts (people, titles, true managers) and never the code. Each comes with an answer key: the topics really discussed, who reports to whom, champions, blockers and decision makers, plus decoys that must not count.

Measure	Set 1, before	Set 2, unseen, after broadening	Set 3, unseen, after broadening
Speakers matched to a person	95 %	100 %	100 %
Known themes, precision / recall	74 % / 83 %	72 % / 67 %	89 % / 85 %
"Who reports to whom" found	0 of 9	1 of 10	**0 of 12**
Champion, blocker, decision maker found	15 of 34	7 of 20	8 of 27
False statements or signals	0	0	0

What this says:

- The meeting patterns never invented a statement.
- They only recognise wordings someone wrote a rule for. Set 1 was used to broaden them, and the gain did not carry over to sets 2 and 3.
- So you chose an opt-in AI pass:
  - Off by default per workspace.
  - It sends only the sentences that name a contact or contain an org phrase, plus the account's contact names, to your own model.
  - Code keeps a finding only if its quote is word for word in the transcript, the names are contacts of the account, any manager is named in the quote, and it is not hypothetical, negated, already true, circular or rejected.
  - Findings are tagged *found by AI*, and a person still accepts each one.
- Its live accuracy will be measured on set 3 once an AI key is saved on the test workspace.

Themes and deal outcomes.

- On the demo, deals whose meetings mention price objections won **16.7 %** of the time, against **57.1 %** for the others: −40.5 points, verified.
- Deals with no meeting are left out, not counted as "no mention".
- This is a screening result:
  - the theme was found on the same meetings;
  - each theme is tested on its own;
  - deals differ in how often, and with whom, they were met.
- The app says so next to the button.

4c. Connecting an AI assistant (M3a, plain language)

What it does. You add one line to Claude Desktop or Claude Code (*Settings → AI assistants (MCP)* shows it). The assistant can then ask DataForce questions about one workspace:

- the overview and pipeline;
- a company or account;
- deals, with filters;
- group comparisons ("does lead source change the win rate?");
- a saved flow;
- meetings, themes and a text search.

Why the answers can be trusted.

- Every number in an answer comes from SQL or the statistics engine, tagged `computed_by: code`.
- Comparisons keep their *verified* / *not verified* label and the reasons, and the assistant is told never to present *not verified* as a fact.

What it cannot do.

- It is read-only, bound to one workspace, and has no network port: the assistant starts DataForce itself on this computer.
- It never receives emails, phone numbers, keys or passwords.

- Everything a customer wrote (notes, transcripts, names, titles, column headers) is marked as data, and the assistant is told not to follow instructions inside it.
- Each call is logged under *Recent calls*, without the data.

4d. The analytics suite (M4, plain language)

Analysis has four tabs.

- **Flows**: the canvas of blocks. M4 adds blocks for duplicate accounts (fuzzy matching, which only proposes groups and never merges records), days since a date, a weighted forecast, month-to-month and quarter-to-quarter growth, web domains, SVM models (a model that draws a boundary between wins and losses), deal velocity, an engagement score, the touchpoint ratio of each rep, and *Score*.
- **Workbooks**: visual canvases of charts (bar, line, area, scatter, pie, donut, heatmap, treemap, funnel), plus number tiles, tables and pivots. Charts are computed on the server; a tile built on a capped source says "computed on the first N rows". **Cross-filtering**: clicking a bar or a slice filters the other tiles. Names in the data are drawn as text, so a name cannot inject markup.
- **Models**: your saved models and *Score now*.
- **Insights**: *Explain*.

Activities and the timeline. On a deal, a rep logs calls, emails and meetings, and moves the deal to a stage on a date (not in the future, not before the deal existed). Each change is recorded. The deal and account pages show one vertical timeline of stage changes, activities, tasks, notes, comments and meeting titles. Flows can read this history, but only the length of each text, never its body.

Saved models and scoring.

- *Save model* appears on a successful model block whose flow is saved. The server re-runs the flow itself and stores its own numbers: the browser sends only a name.
- A saved model is a plain list of numbers, never a program file.
- *Score now* applies it to a CRM table, with the model's own status: **a score is only as verified as its model**. A not-verified model gives scores that say so.
- Rows where a missing value was filled in by an average carry an *imputed* flag. They are not covered by the model's verdict.
- A model that keeps training rows (an RBF SVM, up to 2,000) says so before saving; a linear one keeps none.

Explain. Pick a deal outcome and several attributes; DataForce runs the same statistics as the stats block on each and lists what differs, verified or not. One correction for many comparisons now covers the whole run.

Opt-in AI on deals. *Summarize* and *Suggest tasks* work only after a dialog that shows, without calling the model, what is sent (items, characters, provider, model, calls left today).

- A summary sentence is kept only if it cites a real item of this deal and every number in it appears whole in a cited item. Summaries are not stored.
- A suggested task must quote its source word for word. It stays a *suggestion* until a person accepts it.
- At most 200 calls a day.

The mailbox (Gmail and Outlook).

- You register your own OAuth clients (see `docs/MAILBOX-SETUP.md`); connections are read-only.
- A sync reads the metadata of the last 30 days, at most 500 emails. Only *candidates* are kept: a participant is a CRM contact, or the address is on an account's own domain. Everything else is never stored.
- What is stored: the subject (300 characters), a 200-character snippet, up to 20 addresses.

- Each candidate is scored against open deals, locally: contact email +45, account domain +25, shared words up to +25, recency up to +5. From 40 points an email is *suggested*, with its reasons. You can always link by hand.
- The body is fetched only when an email is linked, cut at 10,000 characters, and becomes an *email* activity on the timeline.
- Disconnecting deletes the tokens and unlinked emails; erasing a contact deletes the stored emails naming them.

What *verified* means here. Unchanged: a model is verified only if, on rows it never saw, it beats the "learnt nothing" baseline with its interval. SVM models follow the same rule. Engagement, velocity and the forecast are labelled as rules of thumb (*heuristic* or *assumptions*), not as verified results.

4e. What the methods review changed (M4)

An independent, read-only reviewer tested the new blocks on synthetic data, then a developer fixed them and the numbers were measured again. The review rated 4 methods high and 6 medium; 7 fixes are done and 3 are partly done.

Method	Problem found	Fix	Measured after
Explain	One correction per attribute let false "verified" results through	One correction across the whole run	Runs with a false verified finding, 5 null attributes, 120 runs: **16 % to 3 %**
Deal velocity	The index flagged many identical deals as stagnant	Within-stage percentile; a stage needs 30 completed stays or its stays are *insufficient*; open stays past the 75th percentile are *stalled*	Flags on a null world with 3 stages: **68 % to 31 %**; identical deals: 0 flags
SVM probability	Probabilities were overconfident with a high setting	Calibration (Platt) fitted on out-of-fold values; the result says the verdict covers ranking and accuracy, not calibration	Calibration error at C = 100: **0.28 to 0.045** (about 9 s more at 20,000 rows)
Scoring	Rows with filled-in values looked verified	A per-row *imputed* flag; those rows are not covered	n/a
Split and leakage	A random split of dated data, or a column known only after close, could be verified	Warnings; a deal-outcome model using a post-outcome column is *not verified*, naming the column	n/a
Touchpoint ratio	A ratio from one email and one meeting looked precise	Needs 10 outreach actions, else empty with a reason	No interval: meetings are not a subset of outreach
Growth (QoQ)	A missing quarter was read as the previous one; tiny or negative bases gave absurd growth	Missing periods count as 0 in a sum; growth is empty on a zero, negative or missing base	n/a
Duplicates	Names sharing a prefix matched; "The Acme" missed "Acme"	Articles stripped, a second blocking key, numbers must agree, a similarity column	n/a

Engagement, forecast	Rules of thumb without a label	Labelled *heuristic* / *assumptions*; accounts with no activity are said to be absent	The 30-day cliff and historical forecast rates are left

Left for later: historical stage-to-win rates for the forecast (it needs a second input), an interval for the touchpoint ratio, and smoothing the engagement cliff.

Honest limits. All experiments are synthetic and show direction and size, not exact production rates.

5. The synthetic demo data

What the generator builds for a realistic small company:

- 30 companies grouping 122 accounts, with invented addresses in real French cities and no SIREN, so they never collide with real firms;
- 755 contacts with org charts;
- 350 deals;
- a year of dated notes and comments;
- **M2b:** half of the accounts lose their manager links in the CRM, so the true tree stays in the answer key for the rules to rediscover; there are 160 meetings (60 % French), varied enough that 85 % of the customer sentences are distinct. They have planted themes (price objection in 65 % of lost-deal meetings vs 20 % of the others), "I report to…" sentences and champion / blocker / decision-maker signals.

Everything is invented and deterministic: the same seed always gives the same data.

Kind	In the demo data
Planted (must be found)	Inbound wins far more often; enterprise deals are slower and larger than SMB deals
Trap (must stay "not verified")	Referral: a high win rate on at most 25 closed deals
Null (must find nothing)	Industry, region and owner. The **noise twin** (seed 1) has no effect at all

Reliability over 100 random datasets per variant (350 deals):

- every planted effect was found every time;
- about 8 % of planted datasets and 4 % of noise datasets show one chance finding.

That is the 5 % false-discovery control doing what it promises. Companies and note dates use separate random streams, so adding them changed no deal and no calibration number.

6. Architecture and structure

```
 browser: React UI (companies, flows, Ask, settings)        CLI  · notebook
     │ JSON on 127.0.0.1 only, request guards
local API (FastAPI)
 ├─ flow engine (pandas, scikit-learn)
 │    └─ sources: CRM · CSV / Excel · PostgreSQL (read-only)
 ├─ companies ── public registry (api.gouv.fr, on click)
 ├─ Ask assistant ── Gemini / Anthropic (key sealed)
 ├─ TamTam connector (OAuth over MCP)
 └─ services layer (workspace-scoped, no SQL strings) ── PostgreSQL
```

Size.

Part	Lines
Backend code	about 33,400
Migrations	1,500
Backend tests	19,400
Frontend (with its tests)	11,000

Main folders.

Folder	Contents
`backend/app/services/`	CRM, companies, files, connections, flows, AI settings, enriched channels
`backend/app/stats/`	Intervals, tests, gates (shared with the flow blocks)
`backend/app/flows/`	Graph contract, engine, one module per block, data sources
`backend/app/registry/`, `backend/app/assistant/`	Public registry client; Ask context, checks and rate limit
`backend/app/connectors/`, `backend/app/llm/`	TamTam OAuth; Gmail and Outlook (M4); Gemini / Anthropic / fake providers
`backend/app/viz/` (M4)	Workbook queries, sources and Explain
`backend/app/flows/scoring.py` (M4)	Applies a saved model's plain parameters to new rows
`backend/app/api/`	Routes, request guards, error mapping, hosting of the interface
`frontend/src/`	Pages; `flows/` (canvas editor); `workbooks/` (ECharts, M4); `models/`, `activity/`, `genai/`, `mailbox/` (M4); `companies/`; `assistant/`; theme

Data model.

Table	Holds
`workspace`, `app_user`	one team (or one demo dataset); its members

`company` (M2a)	name, SIREN, address, activity, size, source; groups accounts
`account`, `contact`, `deal`	the CRM; contacts keep only name, title, reports-to and source
`note`, `comment`	append-only free text
`run`, `llm_call`	history of scans, generations and flow runs (summaries only); AI usage per workspace
`connector_credential`, `oauth_pending`, `contact_channel`	TamTam connection; sign-ins in progress; confirmed emails and mobiles (encrypted)
`uploaded_file`, `data_connection`, `flow` (M2a)	CSV/Excel files; PostgreSQL connections (password encrypted); saved flows
`llm_credential` (M2a)	the workspace's AI provider, model and encrypted key
`title_cache`, `org_proposal` (M2b)	AI title classifications (cached once); suggested links and signals with their evidence and status
`document`, `document_chunk`, `theme`, `theme_assignment` (M2b)	transcript text (cleaned, the file itself is not kept) and speaker turns with exact offsets; themes and which turns carry them
`deal_stage_event`, `activity`, `activity_participant`, `task` (M4)	every stage change with its date; calls, emails and meetings with who took part; tasks, including AI-suggested ones with their quoted evidence
`workbook`, `saved_model` (M4)	visual workbooks (their layout and chart definitions); saved models as plain parameters (never pickle) with their metrics and verdict
`mailbox_connection`, `mailbox_message` (M4)	Gmail / Outlook connections (tokens encrypted); candidate email metadata only, never unrelated mail

Technology.

Choice	Why
Python 3.13, FastAPI, SQLAlchemy 2, PostgreSQL 16	Mature, typed, one database for everything
numpy, scipy, **pandas, scikit-learn, openpyxl** (M2a)	Reference implementations for statistics, tables, models and Excel files
React 19 + Vite + TypeScript, **@xyflow/react** for the canvas (M2a)	Small, fast, no UI kit; the canvas is loaded only when a flow is opened
Apache ECharts 5.6.0 (M4)	The charts of workbooks; pinned and integrity-checked, loaded only on workbook pages, tooltips drawn as text
pypdf in a separate process with a time limit (M2b)	Reads PDF transcripts; a crafted PDF cannot hold the app or exhaust its memory
Fernet encryption	Tokens, keys, passwords and enriched values are unreadable in the database and in backups
Hashed lock files (pip-tools, `package-lock.json`)	Reproducible, tamper-evident installs

7. Security

Principles.

- Local only.
- Another website cannot drive the app: Host allow-list, Origin plus a custom header, strict content-security policy.
- Secrets are encrypted at rest, and never shown, logged or sent to the browser.
- External systems are read-only.
- Text from notes, imports or servers is data, never instructions or HTML.

Reviews (read-only reviewers who must prove each finding).

Review	Scope	Findings	Status
SR4, SR5 (M1)	CRM database, LLM, statistics, demo	7 medium, 11 low, 7 info	all medium fixed
SR6–SR8 (M1b)	local API, TamTam sign-in, frontend	0 medium, 11 low, 11 info	all low fixed
SR9 (M2a)	flows, uploads, PostgreSQL	**1 high**, 7 medium, 6 low, 1 info	high and every medium fixed with tests; 3 low/info accepted with a reason
SR10 (M2a)	companies, registry, AI assistant	1 medium, 5 low, 2 info	all 8 fixed with tests
SR11 (M2a)	frontend, npm supply chain	1 low, 1 info	low fixed
SR12 (M2b)	titles, org proposals, transcripts, PDFs, erasure	**1 high**, 2 medium, 6 low, 3 info	all fixed with tests
SR13 (M2b)	frontend (transcript viewer, suggestions)	3 low, 1 info	all fixed
Methods review (M2b)	accuracy of titles, rules, meeting statements and themes on held-out data	no blocker; 6 should-fix	fixed (see section 4b)
SR14 (M3a)	MCP server and agent tools	2 medium, 4 low, 2 info	all fixed with tests
SR15 (M3a)	opt-in AI pass on meeting sentences	3 medium, 3 low, 1 info	all fixed with tests
SR16	UI identity: profiles, drawer, quality bars	0 high or medium, 3 low, 2 info	3 low fixed; the profile cost accepted
SR17	company-wide assistant context, relaxed AI pass	3 medium, 4 low, 1 info	all 8 fixed with tests
SR18 (M4 phase 1)	activities, timeline, new blocks, workbooks	0 high, 2 medium, 6 low	all 8 fixed with a test each
SR19 (M4 phase 2)	saved models, AI on deals, mailbox	0 high, 3 medium, 10 low	all fixed except the residuals below

M4-METHODS	accuracy of the M4 blocks	4 high, 6 medium (methods)	fixed or partly fixed (section 4e)

M4 and later findings, in plain words.

- **SR17.** Capping a big company's context for the assistant took 21 s per question; now 0.19 s. A "manager named earlier" quote could be a speaker label, a joke or a negation; it must now pass the same guards as the statement, lie within 40 turns, and is cut at 300 characters. Names and titles are wrapped as data, and only code-computed numbers verify an answer.
- **SR18.** Chart tooltips built HTML from data values, so a name like `<img onerror>` could inject markup; they are now drawn as text. Fuzzy duplicate matching could run for minutes before its time limit was checked; it is now bounded. Erasure also masks a contact's name, email and phone inside activities and task evidence.
- **SR19.** A crafted email could freeze a worker for 11 to 15 seconds in the HTML-to-text step; it is now linear (0.1 s). The AI number check was a substring test ("4" passed inside "40000"); numbers now match as whole values.

The M2a high finding. A crafted PostgreSQL query could end the read-only transaction and write to the connected database. The cause was a hand-written statement splitter, fooled by `$$` quoting. The query now runs as a *prepared statement*, which the database server itself refuses to split into two commands, whatever the quoting. Regression tests try every quoting trick; a watchdog also cancels any query that runs past its time limit.

The M2b high finding. Erasing a contact failed when someone's validated manager link came from a proposal naming them, so the right to erasure was blocked. Erasure now clears those links and proposals first.

M3a medium findings fixed.

- **Names reaching the assistant.** Names, titles, labels and flow column headers reached the assistant as ordinary fields, so a customer could plant an instruction in a job title or a CSV header. All customer strings are now declared as data, and flow results never use a customer string as a field name.
- **Flows reaching DataForce's own database.** A flow on a PostgreSQL connection pointed at DataForce's own database could return sealed values and other workspaces' rows. Such connections are now refused when saved and when run, and columns that look like emails, phones or keys are blanked for the assistant.
- **The AI pass: quotes picked to hide meaning.** The model could pick a short quote that hid an "if" or a "not", or that did not name the manager. Checks now run on the whole sentence, and a "reports to" quote must name the manager.
- **The AI pass: daily cap and consent.** Concurrent runs could exceed the daily call cap, now enforced atomically. Changing the AI provider or model kept the switch on; it now switches off, so you consent again for the new provider.

M2b medium findings fixed.

- A crafted PDF page held a worker for minutes, because the time budget was only checked between pages. Extraction now runs in a separate process that is stopped at 30 seconds (with a memory cap where the operating system enforces it).
- An erased person's turns and name stayed in transcripts. They are now redacted.

M2a medium findings fixed.

- A connection whose server was changed kept the old password and would have sent it to the new server. The password must now be re-entered.
- **Heavy inputs could hog the app.** A huge median comparison or a 80,000-column CSV header could run for minutes. Runs now have a concurrency limit, cooperative time limits inside long blocks, and size checks before parsing.
- A pivot could produce a 20 MB answer. Its view is now capped.

- A model tested on its own training data, or statistics on rows duplicated by a join, could show "verified". Both are now caught.
- The assistant's number check could be fooled ("EUR9999", "twelve", numbers taken from notes). Numbers now count only when they appear in code-computed facts; numbers found only in notes are listed apart.

Main things the reviews confirmed hold.

- No route returns or changes another workspace's data.
- Cross-site requests, DNS rebinding and form posts are refused.
- Uploads are accepted only on their own route, with every guard.
- The registry receives your search only on a click (now a guarded POST). It never reads the managers' list, and protected records cannot be stored.
- The AI key never appears in a response, a log or a URL. A website cannot trigger AI calls, and calls are capped at 60 questions per hour per workspace.
- Notes and names cannot break out of the "untrusted text" wrapper sent to the AI.
- The npm dependencies (113 packages) are pinned and integrity-checked, with install scripts disabled.

Accepted residuals (written down):

- **No login until M3b.** Any program on this computer can use the app. The author of a note or activity comes from the request, not from an identity, until then.
- **RBF training rows (SR19-M1).** A saved RBF SVM keeps up to 2,000 training rows as standardised numbers, which survive the erasure of a contact. They are deal and account columns, not contact identifiers, and the Save dialog says so. Deleting the model removes them.
- **Outlook login CSRF on loopback (SR19-B3).** The Outlook callback returns on `localhost`, so the `127.0.0.1` cookie is not sent. The state (single-use, 256-bit, hashed, expiring) and PKCE still bind it; only another local process that learnt the state could race it. Settings shows the connected address.
- **The model cap race (SR19-M5).** Two simultaneous saves can pass the 200-model cap by one.
- **Gmail redirect follows the Host header (SR19-B2).** It is restricted to loopback names on two ports; `docs/MAILBOX-SETUP.md` says which address to open.
- **PostgreSQL.** The database role you connect with remains the real boundary: use a read-only role. `sslmode=require` encrypts but does not verify the server certificate; use `verify-full` for remote databases.
- **Timed-out runs.** A run that times out keeps its worker until the current computation step ends. It holds its slot meanwhile, so runs cannot pile up.
- **Encryption key.** One key, kept in `backend/.env`, with no rotation yet. Losing it means re-entering the AI key and reconnecting TamTam.
- **Unverified assumptions.** TamTam's tool list and prices, and live AI answers, are unverified until you connect real accounts.
- **Mentions in other people's notes.** An erased person's name can remain in other people's notes (append-only).
- **Memory cap on macOS.** The PDF memory cap is not enforced on macOS; the time limit is.
- **Untested on real meetings.** Themes and meeting patterns have been measured on three independently written synthetic sets, never on real transcripts.
- **AI pass, live.** The opt-in AI pass has only run against a fake model so far.
- **Assistants read customer text.** Prompt injection toward a connected assistant cannot be excluded entirely. Its tools are read-only and it never sees keys or contact channels.

8. Verification results

Check	Result
Backend tests	**1,608 pass** (1,417 at M3a; M4, the audit and SR16 to SR19 added the rest), including: CRM, statistics, demo, LLM, TamTam, API, flows, sources, companies, registry, assistant, titles, org proposals, transcripts and PDFs, themes, erasure redaction, **the 14 agent tools, the MCP server (including a real stdio session), the AI pass with a fake model, and one regression test per SR14 and SR15 finding**
Frontend tests	**164 pass**: everything earlier, plus the workbook options (tooltip text, never HTML), column profiles, the timeline logic, the AI assistants panel, the *found by AI* tag and the AI switch notice; type check and build clean
Lint, migrations	ruff clean; models and migrations identical; dev database migrated (indexes added by migration 0008, then the M4 tables)
AI assistant, live	the MCP server started from / the way Claude Desktop starts it; 14 tools listed; overview, comparison, text search and meetings answered on *Test data*, with customer text only in untrusted fields
Independent transcripts	three sets of 15, scored through the running app (section 4b)
M2b against the answer key	titles, rule and meeting links, signals, themes and the price-vs-win-rate test scored through the running app on the seed-42 demo (section 4b)
End to end, through the running app	one flow using all 12 block types on the demo. Inbound and outbound verified, referral not verified (as in M1); the logistic model verified (AUC 0.85); clusters returned; the leakage and join traps caught
Screens	every page opened in a headless browser, in light and dark mode, with no console error; a real registry search for "Dataiku" returned the public record
M4 methods, measured	Explain false verified runs 16 % to 3 %; velocity flags on a null world 68 % to 31 %; SVM calibration error 0.28 to 0.045 (section 4e)
Audit quick wins, measured	workspace delete at 10 times the data 14.6 s to 1.1 s; whole-workspace org proposals at 10 times 72.6 s to 48.6 s; main JS 407 KB to 118 KB, contacts list 418 KB to 55 KB; company context cap 21 s to 0.19 s per question
Held-out set 4	reports-to found 11 of 16, all correct, after the SR17 guards
Notebook	re-run: planted seed 42 finds 5 of 5; noise seed 1 verifies nothing out of 93 comparisons

Caught during the verification itself.

- **M2b themes failed twice before working.** First, small talk and people's names came out as themes. Second, the demo transcripts were too repetitive to judge anything: 25 % distinct sentences, now 85 %. The keyword-first method came out of these measurements.
- **The first "themes vs win rate" table counted deals with no meeting as "not mentioned"**, which biased the comparison. Those deals are now left out.
- **An independent reviewer found the demo accuracy flattering**, and the report now shows the held-out numbers next to the demo ones.
- **Every demo company showed "last interaction: today".** The demo notes had no dates. They now carry a year of synthetic history, which needed a checked "original date" option for imported notes.
- **Dates showed a stray "02:00".** Dates are now shown without a time.
- **The first fix for slot-starved workbooks failed in the browser.** A canvas sends one query per tile at once, so a three-tile workbook got two "another analysis is still running" errors. A tile now waits up to 15 s for a slot.
- **The audit's four "failing tests" were residue of an interrupted run**, not defects; the real fragility (tests counting whole tables) was fixed.
- **The canvas stayed light in dark mode.** It now follows the theme switch.
- **The editor's field names drifted from the API on one point** (a connection's last error). The two now match.

9. How to run it

```
make db-up                                    # local Postgres
make migrate                                  # schema (after the one-time Python setup, see README)
make testdata                                 # the synthetic "Test data" workspace (seed 42)
make ui                                       # then open http://127.0.0.1:8800
backend/.venv/bin/dataforce mcp config --workspace test-data   # snippet for Claude Desktop
```

Try:

Where	What to do
Companies	type "alder", or search the registry for any French company
Analysis	*New flow* → a template, then *Run all* and click a block to see its result
Settings → AI model	paste a Gemini or Anthropic key, *Test*, then *Ask* on a company
Contacts	*Normalize titles*, then *Review titles*
An account	*Suggest links* in the org chart; accept or reject with the evidence; upload a transcript under *Meetings*
Meetings	*Find themes*, then *Test against win rate* on the price theme
Theme	the button in the header
Claude Desktop / Claude Code	paste the snippet from *Settings → AI assistants (MCP)*, then ask "what is the win rate by lead source, and which differences are verified?"
Settings → AI model	tick *Read meeting sentences for org statements*, then *Suggest links* on an account with meetings

A deal (M4)	*Move to stage* with a date, *Log a call, email or meeting*, add a task, then read the timeline; *Summarize with AI* shows what it would send before sending
Analysis → Workbooks (M4)	open *Pipeline by stage*, click a bar to cross-filter, then *Explain*
Analysis → Models (M4)	open *Win probability (sample)* and *Score now* on deals
Settings → Mailbox (M4)	after registering your OAuth clients (`docs/MAILBOX-SETUP.md`), *Connect*, *Sync now*, then link a suggested email on a deal

10. What comes next

Item	Content
Your UI report and the big UI rework	The structural items of the audit go in with it: real paging and windowed tables, no request waterfalls, route-level code splitting, generated API types, a background job for the statistics scan
M3b	Login (email and password, invite-only) and hosting; flow results saved and shared. First, the audit's must-dos: identity and roles with the author set by the server; enforced workspace isolation in the database; shared state for AI budgets and locks; hosting guards; a GDPR erasure registry; versioned encryption keys; a remote story for the MCP server; searchable logs. A scheduler for scoring comes here
Phase-2 leftovers	Historical stage-to-win rates for the forecast; an interval for the touchpoint ratio; smoothing the engagement cliff; a cache for flow sources in workbooks (a tile on a flow re-runs that flow for every query)
Mailbox	Needs your OAuth clients with Google and Microsoft (`docs/MAILBOX-SETUP.md`); then a live trial

Open decisions for you (from the audit).

1. **The embedding stack** (about 150 lines, only tests use it, meant for a semantic search that never shipped): delete it, or build semantic search soon.
2. `GET /titles/preview`, which only tests call: delete it, or keep it for the title editor of the UI rework.
3. `llm_max_calls_per_run`, never enforced: wire it in, or delete it.

Open questions for you.

1. **CRM scope.** After trying Companies, accounts and flows: what is missing?
2. **TamTam.** After connecting, does *Check connection* list the tools we expect, and what are your real credit prices?
3. **AI.** After adding your key, are the Ask answers useful? Should the assistant also be able to *run* a flow for you, for example "pivot my deals by segment"?
4. **Blocks.** Which block do you miss most: export to CSV, time series, a decision tree, or a cohort view?

5. **Real transcripts.** Can you share 5–10 real (anonymised) transcripts? They would give the themes, meeting patterns and the AI pass their first real measurement.
6. **AI pass.** Once your key is saved on *Test data*, the AI pass is measured on set 3. Is a "found by AI" suggestion something you would accept after reading its quote?
7. **Assistant.** After connecting Claude Desktop or Claude Code, which questions did it answer well, and which tool was missing?
8. **M4.** Are the workbook charts, *Explain* and the deal timeline what you pictured? Which chart or block is still missing?
9. **Mailbox.** Can you register the Gmail and Outlook OAuth clients so the linking can be tried on a real mailbox?

Glossary

Term	Meaning
Confidence interval (CI)	The range that most likely contains the true value; narrower with more data
Gate	A minimum-evidence rule; failing one makes a result "not verified"
FDR	The expected share of reported findings that are false, held at 5 %
Train / test split	Keeping some rows aside to judge a model on data it never saw
Baseline	What a model that learnt nothing would score; a model must beat it
AUC	How well a model ranks wins above losses: 0.5 is a coin toss, 1.0 perfect
R^2	Share of the variation a regression explains: 0 is nothing
In-sample / held-out	Measured on the data the method was built with (flattering) / on new data it never saw (honest)
Lexicon	A fixed list of words and phrases per theme, matched by code
Pivot	A table with one dimension in rows, another in columns, and a sum or count in each cell
SIREN	The 9-digit French company identifier
Prepared statement	A query the database checks as one single command before running it
MCP	Model Context Protocol: the standard way AI agents call tools
Workbook	A canvas of charts and tiles in the Analysis tab, computed on the server
Cross-filtering	Clicking a mark in one chart filters the other tiles of the canvas
Explain	The workbook tool that compares a measure across the groups of one or more attributes, verified findings first, with the usual verdicts
Saved model	A fitted model kept as plain numbers (never a program file), so it can be reused
Scoring	Applying a saved model to rows; a score is only as verified as its model

Platt calibration	Turning a model's raw output into a probability; fitted on data the model did not train on
Out-of-fold	Predictions made on rows that were left out of that fold's training, so they are honest
BH / false discovery	Benjamini-Hochberg: the correction that keeps the expected share of false findings at 5 % across many comparisons
PKCE	A one-time proof that the program finishing a sign-in is the one that started it
OAuth client	The registration of an app with Google or Microsoft (an id and a secret) that lets you sign in and grant read-only access